Insulin-Like Growth Factor 1 Activates PI3K to Induce AKT Phosphorylation and Suppress Apoptosis

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Page 1 — Abstract & Introduction

Survival factors protect neuronal and epithelial cells against stress-induced programmed cell death.

Page 2 — Early Transduction

Binding of IGF-1 to its receptor activates PI3K lipid kinase, generating PIP3 at the plasma membrane.

Page 3 — Kinase Activation

Membrane PIP3 recruits and activates AKT via phosphorylation at Thr308 by PDK1 and Ser473 by mTORC2.

Page 4 — Substrate Phosphorylation

Activated AKT directly phosphorylates BAD at Ser136, promoting BAD sequestration by 14-3-3 proteins.

Page 5 — Discussion & Conclusions

Sequestration of phosphorylated BAD inhibits apoptosis and promotes sustained cell survival. Thus, IGF-1 suppresses apoptosis through PI3K and AKT signaling.